

Software Development Life Cycle -- Guided Notes ANSWER KEY

Unit tp-4.1 | CompTIA Network+ N10-009 Obj. FC0-U71 4.1 | N10-008 FC0-U71 4.1

For instructor use / distribute after students complete the notes.

Question 1

The SDLC (Software Development Life Cycle) is a structured process for planning, creating, and maintaining software. The six main phases are: _____, _____, _____, _____, _____, and _____.

Answer: Planning, requirements analysis, design, development (coding), testing, deployment (and maintenance).

Question 2

In the _____ phase, stakeholders define what the software must do. These are captured as functional requirements (what it _____) and non-functional requirements (how well it _____, such as performance and security). Unclear requirements at this stage lead to _____ later.

Answer: Requirements analysis; does; performs; rework (costly changes).

Question 3

In the _____ phase, developers write code based on the design documents. Code is typically managed in a _____ control system like Git. After coding, the _____ phase verifies the software meets requirements and finds defects before release.

Answer: Development (implementation); version (source); testing (QA).

Question 4

The Waterfall model is _____ -- each phase must fully complete before the next begins. It works well when requirements are _____ and unlikely to change. The main risk is that mistakes discovered _____ in the process are very expensive to fix.

Answer: Sequential (linear); stable (well-defined); late.

Question 5

Agile is an iterative SDLC methodology where work is done in short cycles called _____. At the end of each cycle, a working _____ is delivered and requirements can be _____. Agile prioritizes collaboration with _____ over rigid planning.

Answer: Sprints (iterations); product increment; adjusted (changed); customers (stakeholders).

Question 6

DevOps combines _____ and _____ teams to automate and integrate the build, test, and deployment pipeline. CI/CD stands for _____ and _____. With CI, every code commit triggers _____ automated tests to catch bugs immediately.

Answer: Development; operations; Continuous Integration; Continuous Delivery/Deployment; automated.

Question 7

Version control (source control) allows developers to track _____ over time. A _____ is a parallel line of development used to add a feature or fix a bug without disturbing the main code. After testing, the branch is _____ back into the main codebase. Git is the most widely used _____ version control system.

Answer: Changes; branch; merged; distributed.

Question 8 -- Real World Application

REAL WORLD: A startup uses Waterfall to build their mobile app over 18 months. At the end of development, they discover that users want a completely different onboarding flow than what was designed at the start. All the code must be rewritten. What SDLC methodology would have prevented this and why?

Answer: Agile would have prevented this. In Agile, the team works in short sprints (2-4 weeks) and delivers a working increment at the end of each sprint. The onboarding flow would have been designed, coded, and shown to real users in sprint 1 or 2. User feedback would have been collected early, when changes are cheap (a few days of work). By delivering working software frequently, Agile surfaces misaligned assumptions before 18 months of work has been invested. In Waterfall, requirements are locked in up front and feedback does not happen until the final product is delivered -- making late-stage user feedback catastrophically expensive to act on. Agile's core advantage is embracing change throughout the process rather than resisting it.